

ELEC5305 PROJECT PROPOSAL

Evaluation of VAD-Guided Wiener Filtering for Speech Enhancement Using Real Recordings in Everyday Noise

Speech and Audio Processing

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GitHub Project Link	https://github.com/yche0441/elec5305-project-550889472

1. Project Overview

Speech recorded in everyday environments often contains fan noise, traffic noise, and other background sounds. These sounds can make speech less clear and may also reduce the reliability of later tasks such as speech recognition. This project will examine whether a voice activity detection (VAD) stage can improve a Wiener speech enhancement system when the noise changes over time. VAD is a method that identifies frames that are likely to contain speech. A Wiener filter then applies a frequency-dependent gain to reduce components that are more likely to be noise.

The research question is: How much does VAD-guided noise updating improve noise reduction and speech preservation compared with a Wiener filter that uses one fixed noise estimate when both systems are tested on real recordings made in everyday noisy environments? The project will implement both methods in MATLAB and compare them under the same recording conditions.

2. Background and Motivation

Single-channel speech enhancement is useful when only one microphone recording is available. A common approach is to divide the recording into short overlapping frames and analyse each frame in the frequency domain. The short-time Fourier transform (STFT) provides this time-frequency representation. A Wiener filter can then reduce frequency components according to the estimated balance between speech and noise. Scalart and Filho described a practical approach based on estimating the prior signal-to-noise ratio, while Ephraim and Malah showed how short-time spectral estimation can reduce noise while limiting speech distortion [1], [2].

The quality of the noise estimate is important. A fixed estimate may work for steady noise, but it may become inaccurate when the background changes. VAD provides a way to identify likely non-speech frames and update the noise estimate during those frames. Sohn, Kim, and Sung

demonstrated the value of statistical VAD for speech processing in noisy conditions [3]. Cohen and Berdugo also showed that adaptive noise estimation is useful when noise is not stationary [4].

This topic was selected because it has a clear comparison, can be implemented with techniques covered in ELEC5305, and can be demonstrated using sound examples and spectrograms. The project will use only real acoustic recordings. It will not generate artificial noise in MATLAB, synthesise noisy speech, or digitally mix clean speech with a noise file.

3. Proposed Methodology

The recordings will be made by the student using the same recording device and a similar microphone distance where practical. The same short passage will be spoken in three conditions: a quiet indoor room, a room with a real electric fan operating, and an indoor location where real traffic noise is audible through an open window. Each noisy recording will include approximately five to ten seconds of noise without speech. These sections will provide initial noise information. Recording details, including device, sampling rate, distance, and environment, will be documented. The recordings will be converted to mono if required and resampled to a common sampling rate before analysis.

MATLAB will be used for all signal processing. Each recording will first be inspected using a waveform, power spectrum, and spectrogram. The signal will then be divided into overlapping short frames with a Hamming window, and an STFT will be calculated. Three outputs will be retained for comparison: the original unprocessed recording, a baseline Wiener-filter output, and a VAD-guided Wiener-filter output.

For the baseline system, the noise power spectrum will be estimated from the opening noise-only section and then held fixed. For the proposed system, a frame-energy VAD will identify likely speech and non-speech frames. The noise estimate will be updated only during likely non-speech frames. Both systems will use a Wiener gain derived from the estimated speech and noise power. A lower gain limit will be tested to reduce strong changes between neighbouring frames, which can otherwise create unnatural residual sounds. The enhanced waveform will be reconstructed with the inverse STFT and overlap-add processing.

The comparison will use measures that can be calculated without claiming that two separately recorded signals are identical. Noise attenuation will be measured from manually checked non-speech intervals. An estimated speech-to-noise ratio will be calculated from labelled speech and non-speech frames. Speech preservation will be examined by comparing speech-frame energy and spectral shape before and after processing. Processing time will also be recorded. Waveforms, spectrograms, numerical tables, and short listening examples will be used together because one measure alone cannot describe all changes in speech quality. The limitations of using a small student-recorded dataset and approximate noise labels will be stated clearly.

4. Expected Outcomes

The expected deliverables are a documented MATLAB implementation, the real recordings used for testing where sharing is appropriate, processed audio examples, plots, a comparison table, a written report, and a short demonstration video. The project does not assume that the VAD-guided method will always perform better. It will report whether the adaptive noise update provides a measurable benefit for steady fan noise and changing traffic noise, and it will also identify cases where speech is weakened or residual noise remains. The final repository will include instructions that allow the code and examples to be reproduced.

5. Timeline (Weeks 1-13)

Weeks	Planned Task
1-2	Confirm the research question, project scope, and GitHub structure.
3-5	Review relevant literature and collect real speech and noise recordings.
6-7	Implement STFT analysis and the fixed-noise Wiener baseline.
8-9	Implement energy-based VAD and adaptive noise updating.
10-11	Run tests, compare measures, and examine failure cases.
12-13	Complete the report, code documentation, GitHub Pages site, and video.

6. References

- [1] P. Scalart and J. V. Filho, "Speech enhancement based on a priori signal to noise estimation," in Proc. IEEE Int. Conf. Acoustics, Speech, and Signal Processing (ICASSP), vol. 2, 1996, pp. 629-632, doi: 10.1109/ICASSP.1996.543199.
- [2] Y. Ephraim and D. Malah, "Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator," IEEE Trans. Acoustics, Speech, and Signal Processing, vol. 32, no. 6, pp. 1109-1121, Dec. 1984, doi: 10.1109/TASSP.1984.1164453.
- [3] J. Sohn, N. S. Kim, and W. Sung, "A statistical model-based voice activity detection," IEEE Signal Processing Letters, vol. 6, no. 1, pp. 1-3, Jan. 1999, doi: 10.1109/97.736233.
- [4] I. Cohen and B. Berdugo, "Noise estimation by minima controlled recursive averaging for robust speech enhancement," IEEE Signal Processing Letters, vol. 9, no. 1, pp. 12-15, Jan. 2002, doi: 10.1109/97.988717.

7. Appendix (Optional)

The GitHub Pages site will be published at <https://yche0441.github.io/elec5305-project-550889472/> after Pages is enabled. Preliminary plots and audio demonstrations will be added after the real recordings have been collected and the MATLAB implementation has been tested.